# SANIKA PRASAD NAIK

*Data Scientist* | *ML Research* | *Python* | *SQL* | *PyTorch* | *Explainable AI* | *Agentic AI* | *RAG*

naiksanika70@gmail.com | Birmingham, UK

linkedin.com/in/sanikanaik1308 | github.com/sanikanaik456 | portfolio-aa5vrerkprnrgx3drn6adb.streamlit.app

**Graduate Route visa — full right to work in the UK for 2 years from December 2026, no sponsorship required**

## PROFILE

MSc Data Science graduate (University of Birmingham, 2026) with specialist experience in credit risk modelling, fraud analytics, bias detection and generative AI. Built production-ready RAG pipelines and agentic AI systems, and authored a survey paper on Agentic AI Systems covering LangChain, RAG architectures and responsible AI deployment. Genuine passion for using data and AI to make financial services fairer and more transparent for everyone.

## TECHNICAL PROJECTS

**Explainable Deep Learning for Credit Risk Prediction** | *Python, PyTorch, SHAP, SQL, Pandas, scikit-learn, Power BI, Streamlit*

*MSc Dissertation, University of Birmingham, 2026*

- Improved credit default prediction accuracy from 82.9% to 83.8% by implementing group-specific decision thresholds that equalised false positive rates across demographic groups.
- Reduced demographic bias by 58% (from 36% to 15% false positive rate for under-35 borrowers) by applying proportional sampling and threshold calibration techniques across 150,000 financial records — addressing population bias in credit decisioning.
- Achieved AUC 0.850 on credit default prediction by training PyTorch neural networks with 5-fold cross-validation (0.837 ±0.003), benchmarked against logistic regression and XGBoost baselines.
- Discovered that SHAP and Integrated Gradients produce conflicting feature importance rankings for the same model due to differing reference baselines — a novel finding with direct implications for explainable AI in regulated financial services.
- Built a six-tab Streamlit Decision Support System enabling live predictions, SHAP explainability, counterfactual advice and fairness audit — making complex AI outputs accessible to non-technical banking officers.

**DocuChat — RAG Pipeline and Generative AI Application** | *Python, Streamlit, Google Gemini API, FAISS, LangChain-style RAG*

- Reduced LLM hallucination rate to zero by implementing retrieval-augmented generation with FAISS vector search, grounding all outputs in retrieved document context.
- Deployed a production-ready Streamlit application end to end, demonstrating ability to take AI solutions from concept through deployment. GitHub: github.com/sanikanaik456/docuchat

**Customer Churn Prediction** | *Python, scikit-learn, SQL, Pandas*

- Improved customer retention insight by achieving ROC-AUC 0.842 on churn prediction, identifying top behavioural drivers and translating findings into actionable recommendations for product teams.

**Agentic AI Systems — Survey Paper** | *LangChain, LangGraph, RAG, Google ADK, Multi-Agent Architectures*

- Authored a 28-page survey covering RAG architectures, LangChain and Google ADK agent frameworks, LLM evaluation methodologies and responsible AI deployment — covering the main frameworks and evaluation approaches in current agentic AI research.

## EXPERIENCE

**Data Analyst Intern (Remote), Faceify Labs (Formir Technologies Pvt Ltd)** *Jan to Jul 2026*

- Increased analytical efficiency by building Python data pipelines that automated recurring data processing tasks, reducing manual analysis time across the team.
- Improved stakeholder decision-making by developing Power BI dashboards tracking key performance metrics, enabling data-driven decisions for product and leadership teams.
- Reduced model deployment time by containerising a classification model using Docker, applying CI/CD-adjacent engineering practices for production deployment.

**Data Analysis Project Assistant (Remote), Private Client** *2023 to 2025*

- Delivered measurable analytical value across 14 independent data science projects, consistently translating complex data into actionable business recommendations for diverse commercial clients.

## EDUCATION

**MSc Data Science, University of Birmingham, United Kingdom** *2025 to 2026*

- Passed with Merit. Dissertation: Explainable Deep Learning for Credit Risk Prediction (credit risk modelling, bias detection, proportional sampling, RAG). Survey paper: Agentic AI Systems (LangChain, Google ADK, RAG architectures). Key modules: Advanced AI, SQL and Databases, Large Language Models, Python.

**B.E. AI and Data Science, Savitribai Phule Pune University (SPPU), Pune** *2021 to 2025*

- First Class with Distinction.

## SKILLS

- **Languages and Databases:** Python, SQL (Advanced), MySQL, PostgreSQL
- **ML and AI:** Credit Risk Modelling, Statistical Modelling, Machine Learning, Deep Learning, Generative AI, RAG Pipelines, Agentic AI, LangChain (conceptual), Google ADK (conceptual), LLMs, Prompt Engineering, Explainable AI (SHAP), Fraud Analytics, Bias Detection, A/B Testing; PyTorch, XGBoost, scikit-learn, Pandas, NumPy, Spark (familiar)
- **Tools:** Streamlit, Docker, Git/GitHub, Jupyter, Power BI, Google Cloud (familiar), Excel

## CERTIFICATIONS

HackerRank SQL (Advanced) Certification, 2026. Deloitte Data Analytics Job Simulation (Forage), 2026. National-level chess player (Maharashtra state qualifier, 2018). Languages: English (fluent), Hindi (fluent), Marathi (fluent).